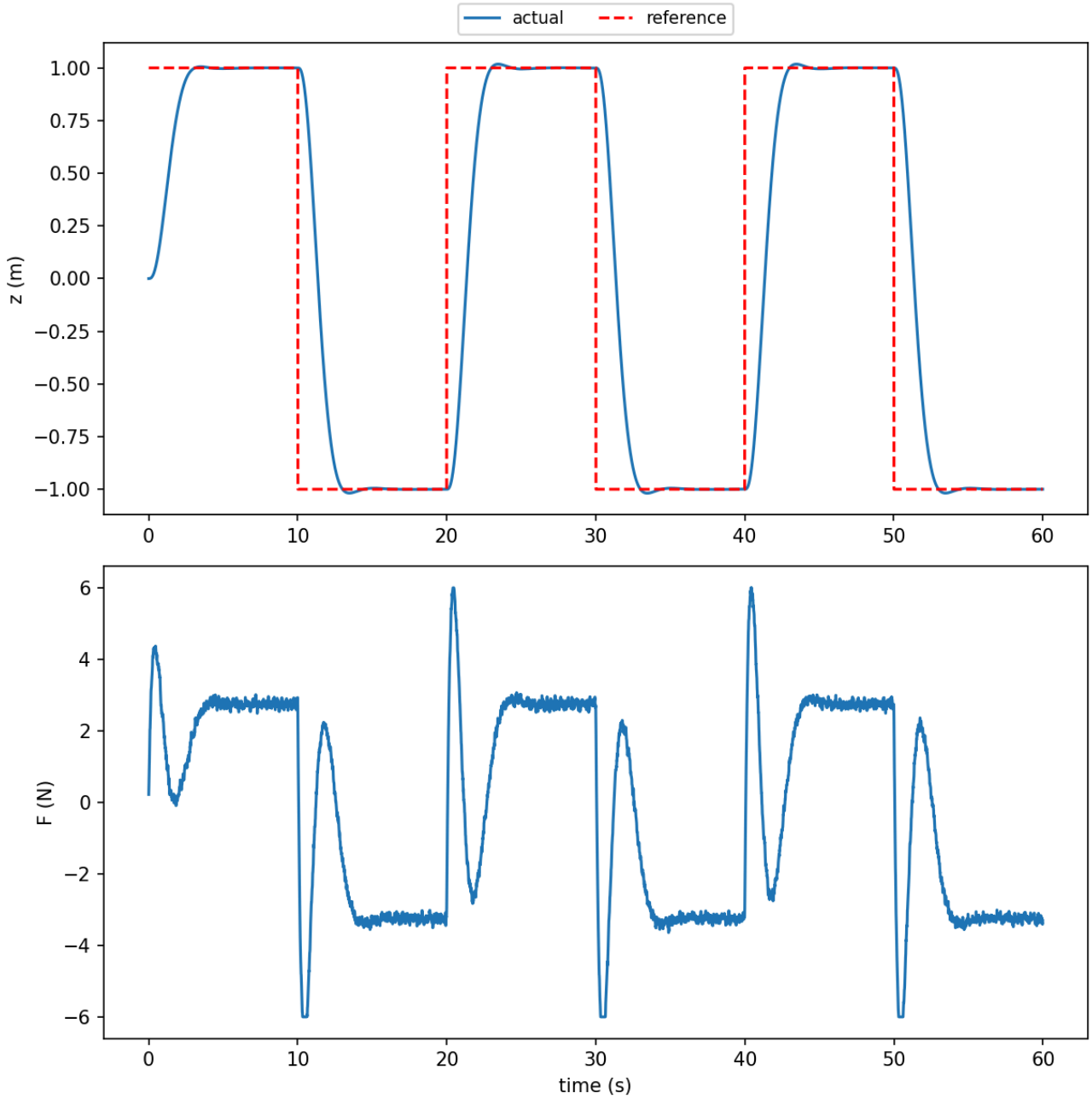
LQR HW

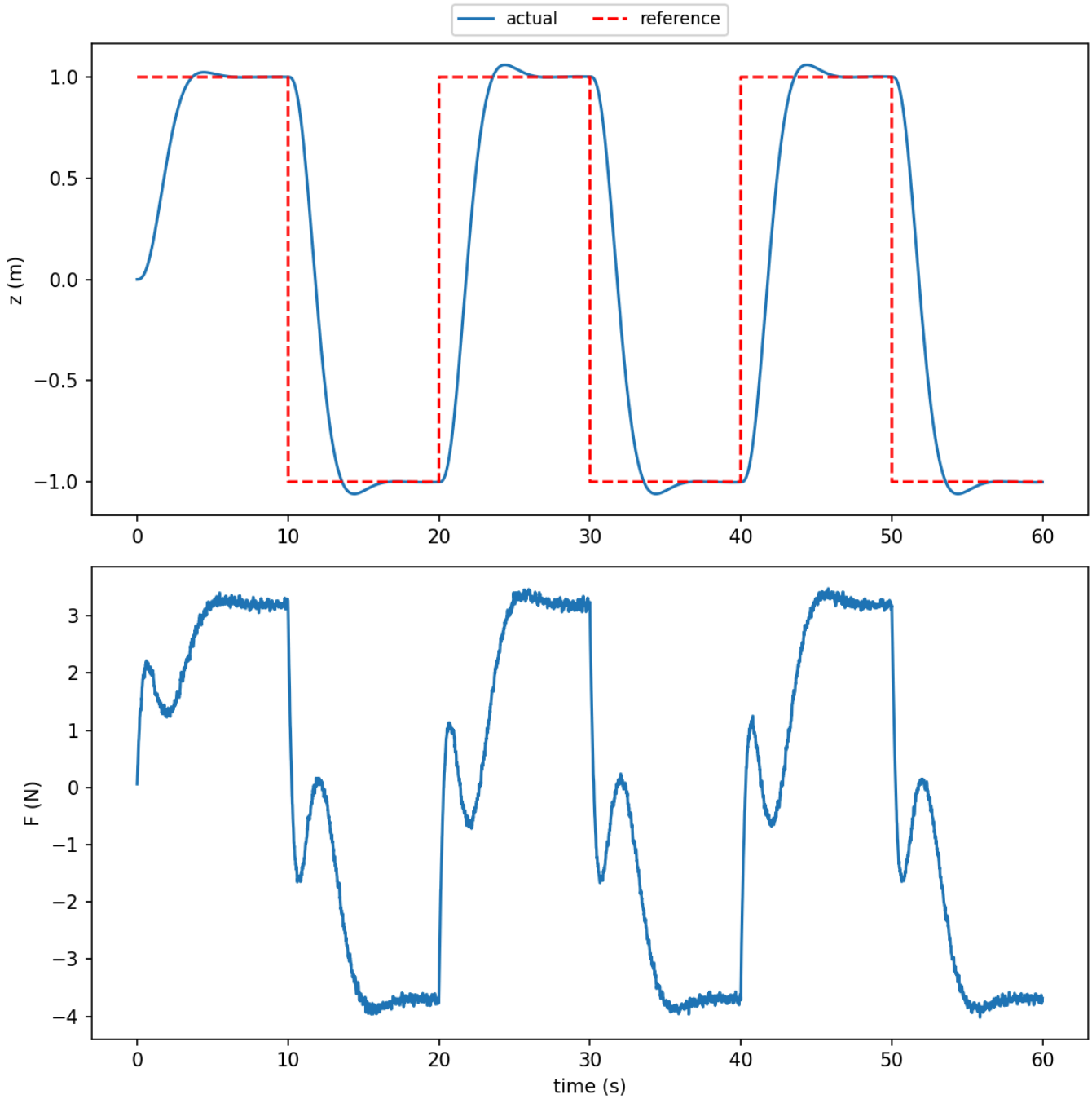
Mass



1.

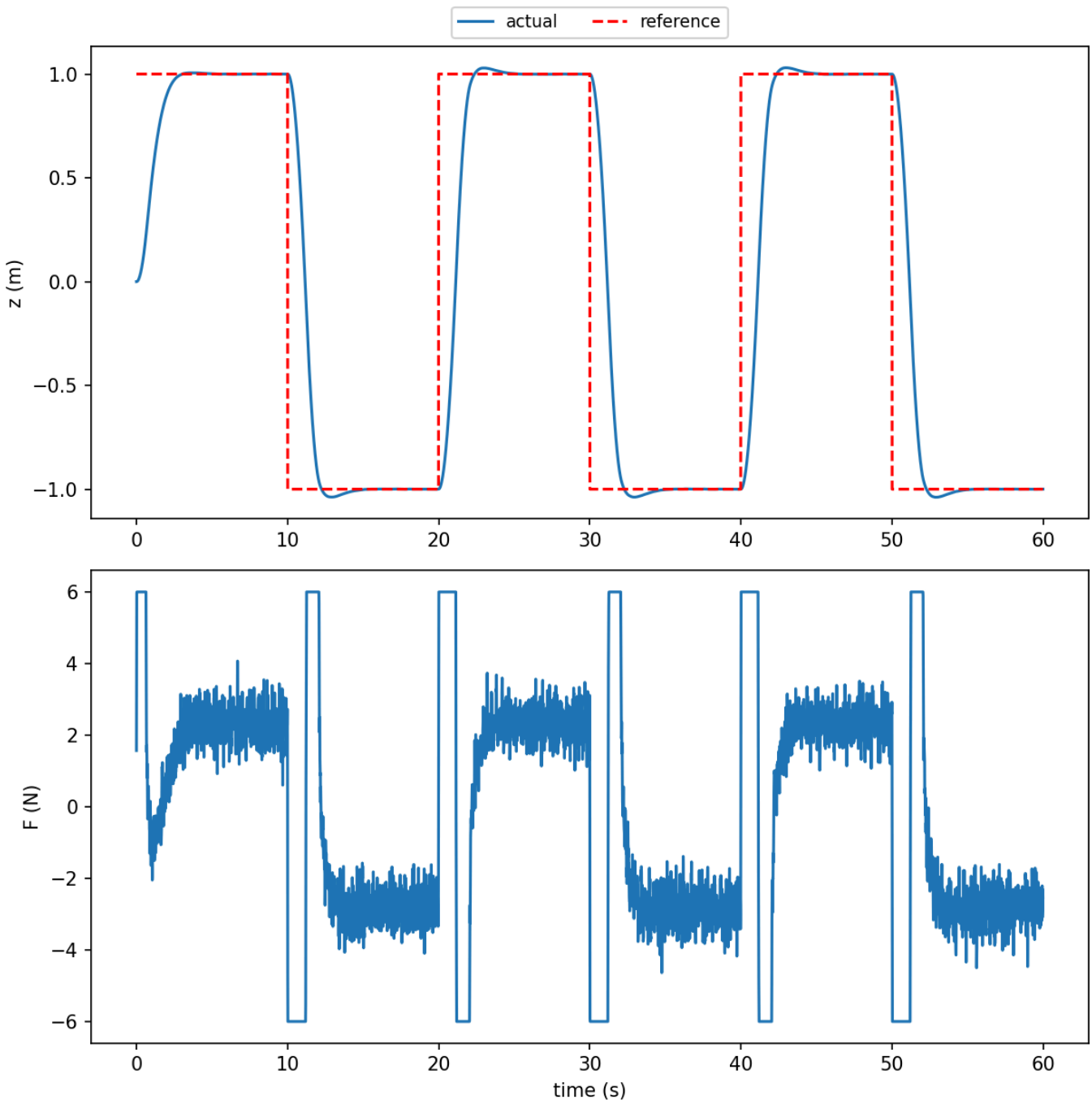
```
eigs = array([-1.2591+1.2947j, -1.2591-1.2947j, -1.6223+0.j    ])
```

2a)




```
eigs = array([-0.6512+1.1425j, -0.6512-1.1425j, -0.9677+0.j    ])
```

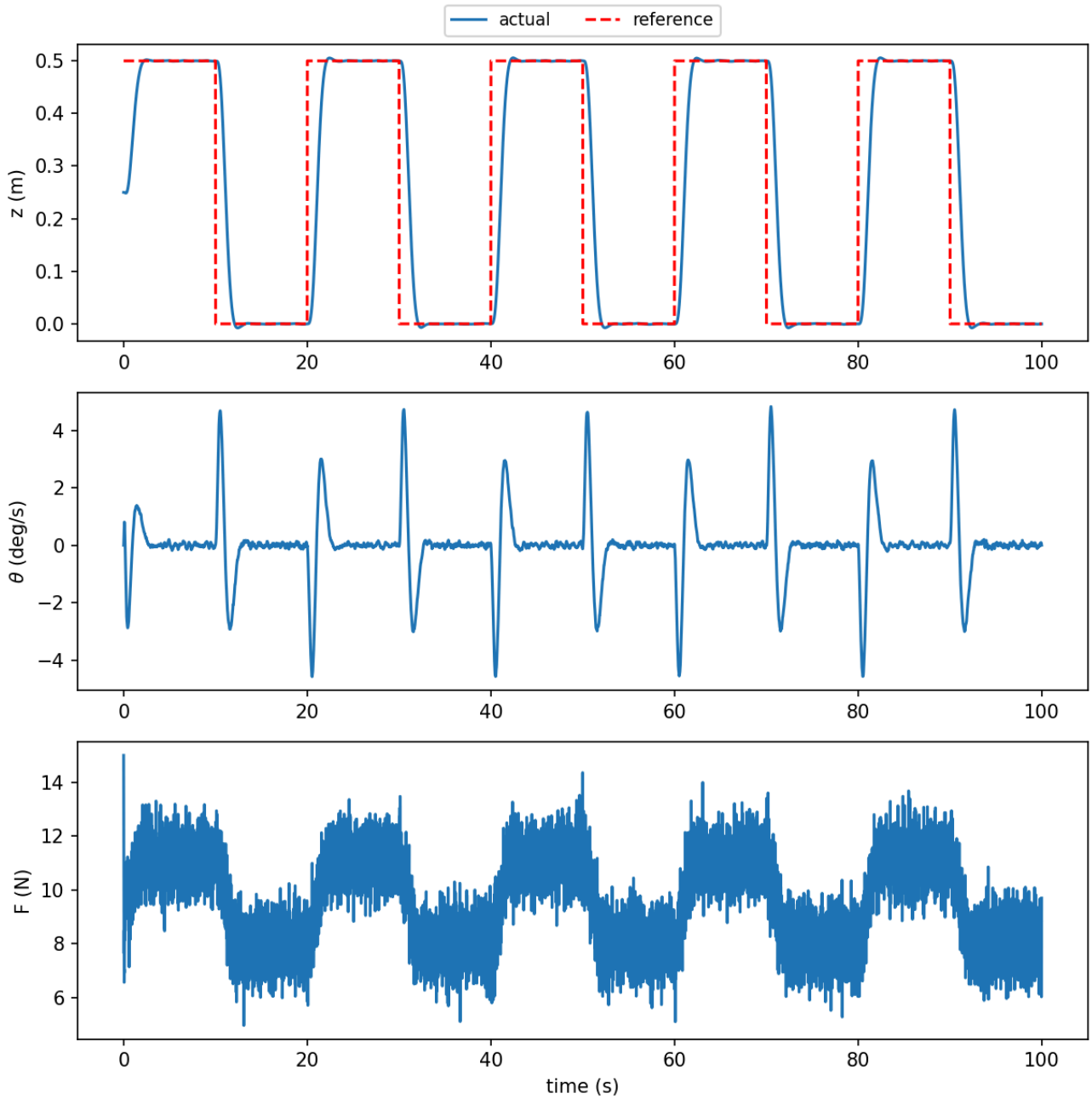
2b)




```
eigs = array([-22.9268+0.j    ,  -1.3159+0.8374j,  -1.3159-0.8374j])
```

3)Yes the pole values make sense. Higher pole values mean faster response.

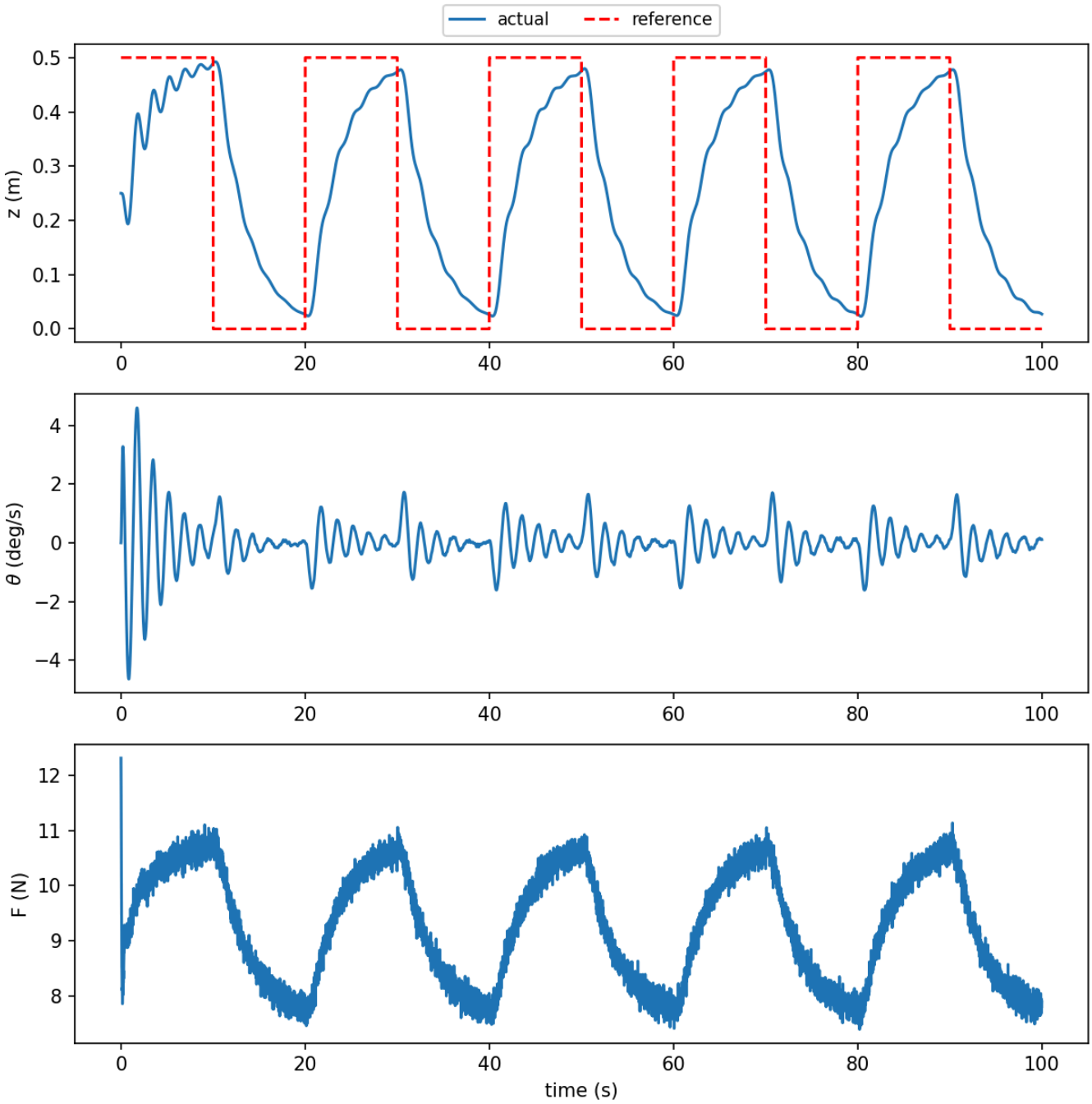
Blockbeam



1.

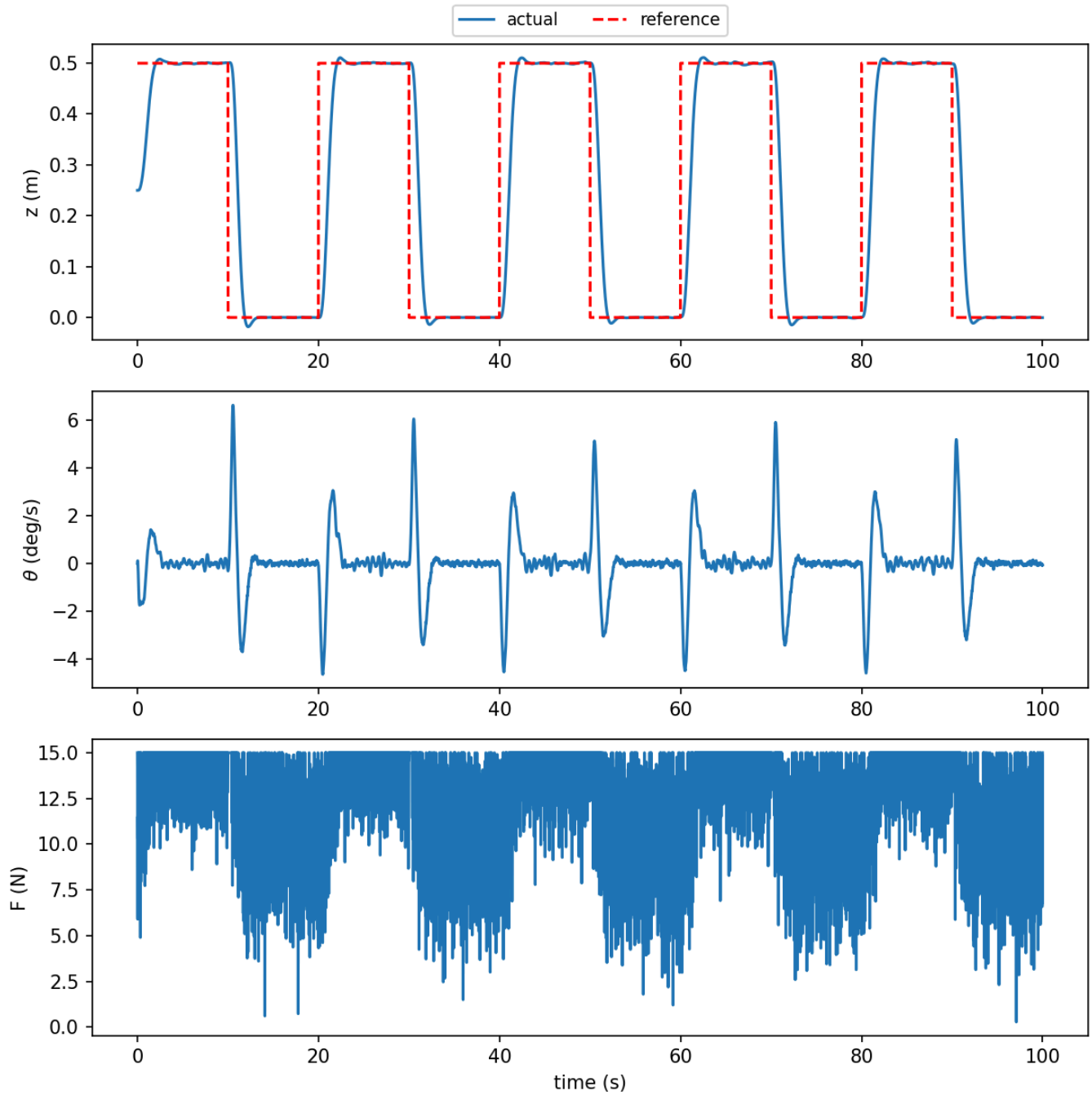
```
eigs: [-32.4762+0.j       -2.1175+1.1754j  -2.1175-1.1754j  -2.0515+2.6846j
  -2.0515-2.6846j]
```

2a)




```
eigs: [-0.3052+3.6485j -0.3052-3.6485j -3.9315+0.j     -3.4038+0.j
 -0.313 +0.j    ]
```

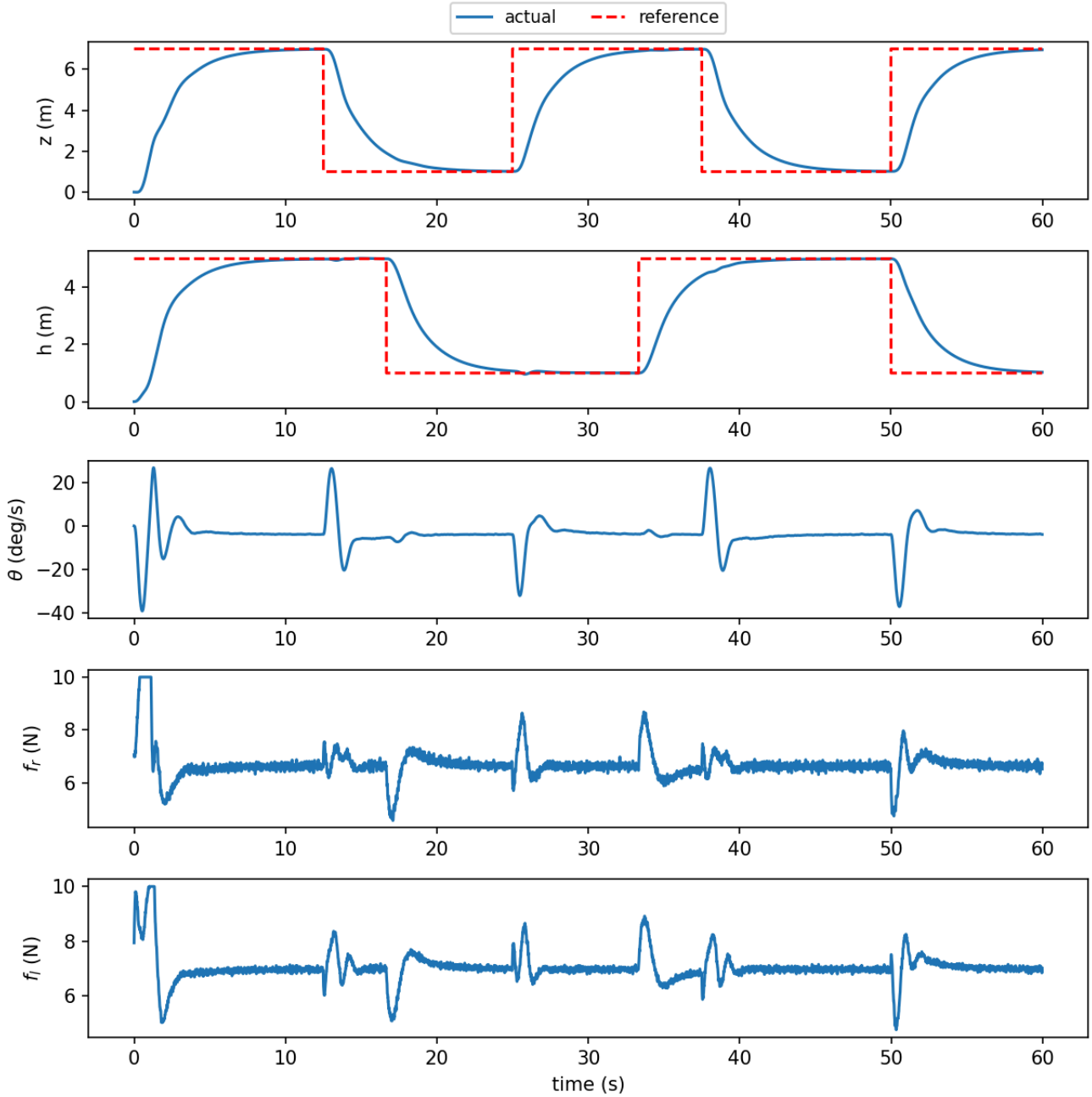
2b)




```
eigs: [-102.7059+0.j        -2.124 +1.2024j   -2.124 -1.2024j   -2.0595+2.6452j
   -2.0595-2.6452j]
```

3)Yes the pole values make sense. Smaller pole values correspond to slower response. It is interesting that 2a sort of moves the poles closer together.

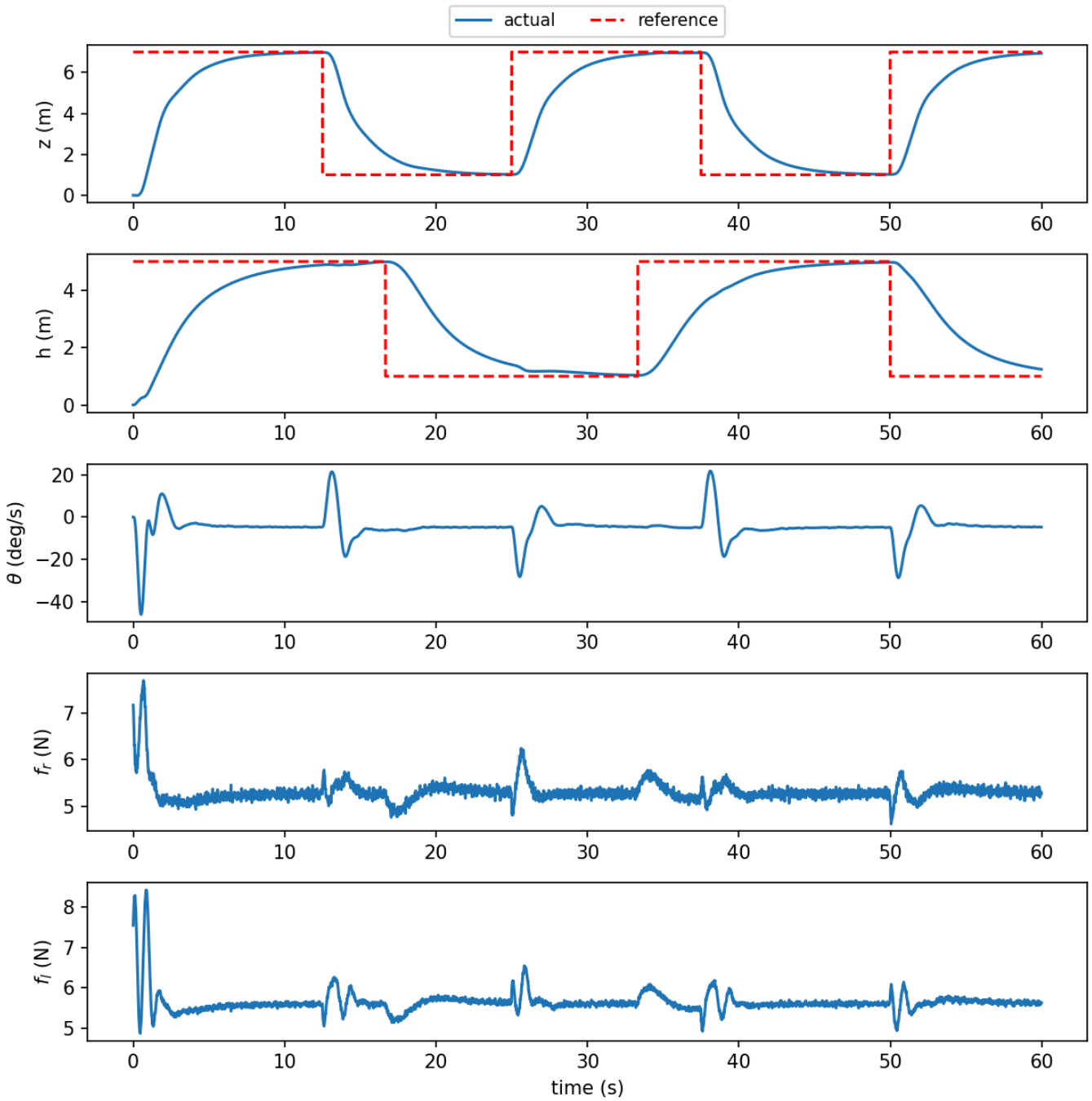
VTOL



1.

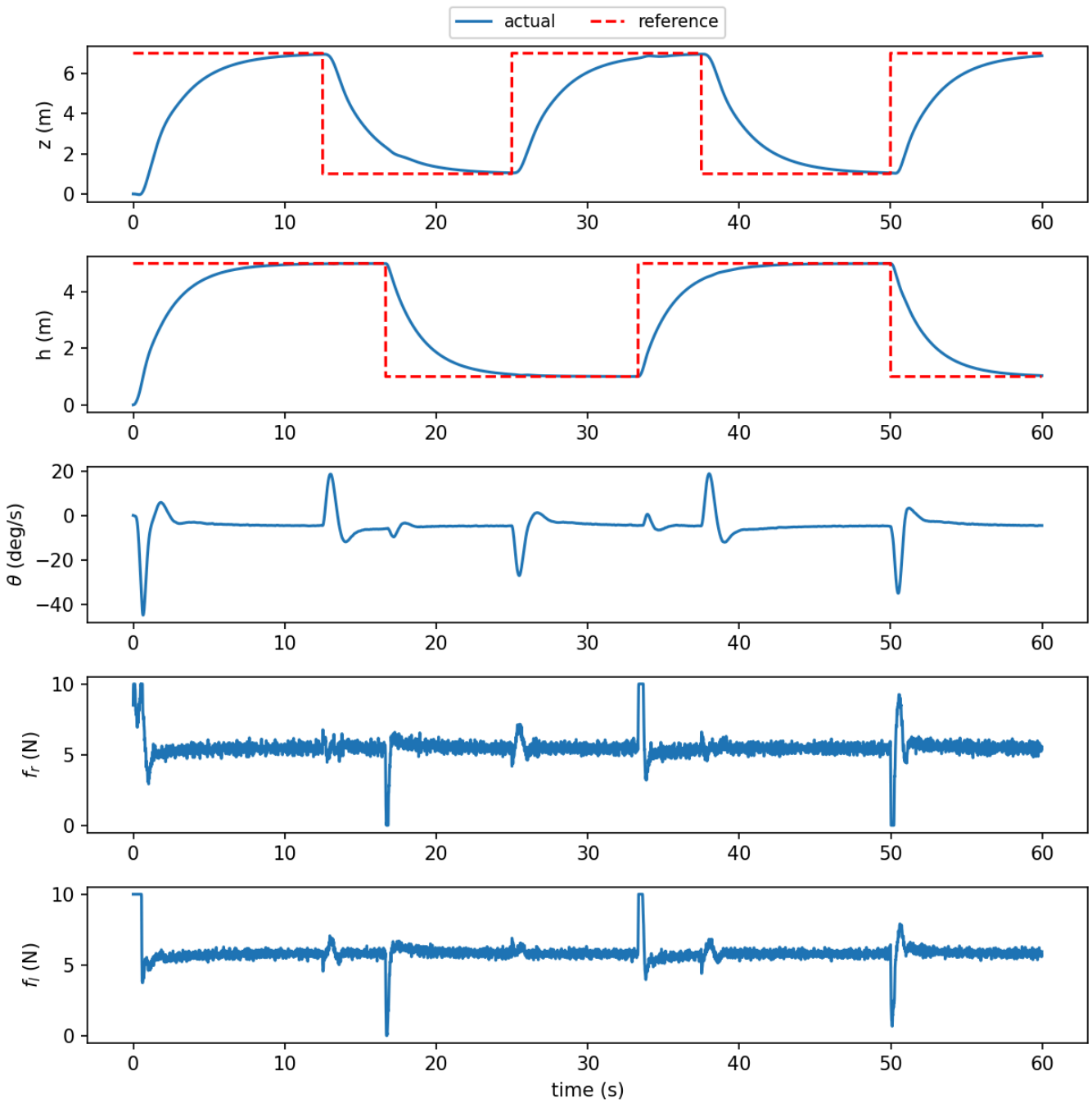
```
Closed loop eigenvalues: [-11.5267+0.j      -2.122 +3.3441j  -2.122 -3.3441j  -3.9307+0.j
  -1.6598+0.j      -1.4876+1.8091j  -1.4876-1.8091j  -1.4884+0.j    ]
```

2a)




```
Closed loop eigenvalues: [-1.8558+3.3396j -1.8558-3.3396j -4.3413+1.0182j -4.3413-1.0182j
 -1.6593+0.j      -0.5548+0.8577j -0.5548-0.8577j -0.9258+0.j     ]
```

2b)




```
Closed loop eigenvalues: [-49.0172+0.j      -2.0938+2.9999j  -2.0938-2.9999j  -3.8867+0.j
  -1.1287+0.j      -7.0831+6.6364j  -7.0831-6.6364j  -1.0376+0.j    ]
```

3)The eigenvalues mostly make sense, but it's interesting that there isn't really much better response in 2b than in 1. It's pretty difficult to make the gains higher without instability.